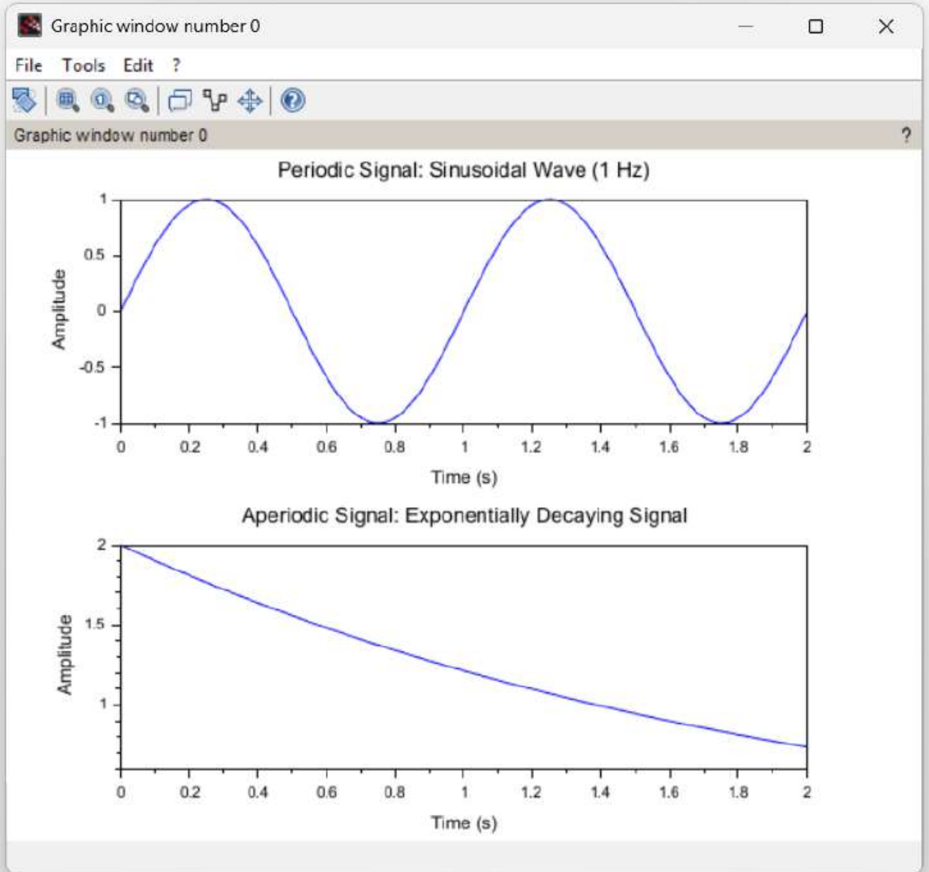
EXP_2 2.1 SS.sci (E:\EXP_2 2.1 SS.sci) - SciNotes

EXP_2 2.1 SS.sci

```
// Test Case: Generation of Periodic and Aperiodic Signals
// Parameters for the periodic signal (Sinusoidal)
f_periodic = 1;        // Frequency in Hz
A_periodic = 1;        // Amplitude
t_periodic = 0:0.01:2; // Time vector for 0 to 2 seconds
// Generate the periodic signal
x_periodic = A_periodic * sin(2 * %pi * f_periodic * t_periodic);
// Plot the periodic signal
clf; // Clear the current figure
subplot(2,1,1);        // Subplot 1 for the periodic signal
plot(t_periodic, x_periodic);
 xlabel('Time (s)');
 ylabel('Amplitude');
 title('Periodic Signal: Sinusoidal Wave (1 Hz)');
 // Parameters for the aperiodic signal (Exponentially Decaying)
 a_aperiodic = 0.5;     // Base of the exponential decay
 A_aperiodic = 2;       // Amplitude
 t_aperiodic = 0:0.01:2; // Time vector for 0 to 2 seconds
 // Generate the aperiodic signal
 x_aperiodic = A_aperiodic * exp(-a_aperiodic * t_aperiodic);
  // Plot the aperiodic signal
  subplot(2,1,2);        // Subplot 2 for the aperiodic signal
  plot(t_aperiodic, x_aperiodic);
  xlabel('Time (s)');
  ylabel('Amplitude');
   title('Aperiodic Signal: Exponentially Decaying Signal');
```



EXP_2.2 SS.sci (E:\EXP_2.2 SS.sci) - SciNotes

EXP_2.2 SS.sci

```
t = -1:0.001:1;
x = exp(-abs(t)).*sin(4*%pi*t) + 0.3;
// Reverse signal
x_rev = interpl(t, x, -t, 'linear');
// Even and odd parts
xe = 0.5 * (x + x_rev);
xo = 0.5 * (x - x_rev);
clf;
plot(t, x, 'k');
plot(t, xe, 'b--');
plot(t, xo, 'r:');
legend("x(t)", "Even Part", "Odd Part");
xtitle("Even-Odd Decomposition", "t", "Amplitude");
```